6-

```
postgres=# CREATE VIEW student_subject AS
                     SELECT e_name ,subject.sub_name
                  FROM student
                    JOIN stu_sub ON student.stu_id=stu_sub.stu_id
                      JOIN subject ON stu_sub.sub_id=subject.sub_id;
CREATE VIEW
postgres=#
```

7-

```
postgres=# CREATE VIEW track_namee AS
                 SELECT name ,subject.sub_name
            FROM track
            JOIN track_sub ON track.track_id=track_sub.track_id
                    JOIN subject ON track_sub.sub_id=subject.sub_id;
CREATE VIEW
postgres=#
```

lab3

1-

```
postgres=# SELECT
    s.sub_id AS subject_id,
    s.sub_name AS subject_name,
    s.max_score
FROM
    subject s
WHERE
    NOT EXISTS (
        SELECT 1
        FROM track_sub ts
        WHERE ts.sub_id = sub_id
    )
ORDER BY
    s.sub_name;
 subject_id |   subject_name   | max_score
------------+------------------+-----------
          2 | Calculus I       |       100
          1 | Database Systems |       100
          3 | Data Structures  |       100
(3 rows)

postgres=#
```

2-

```
postgres=# SELECT
    e_name AS student_name,
    EXTRACT(YEAR FROM AGE(birth_date)) AS age
FROM
    student
ORDER BY
    e_name;
 student_name | age
--------------+-----
 arwa         |  25
 aya          |  24
 menna        |  23
(3 rows)

postgres=#
```

3-

```
postgres=# SELECT
    s.e_name AS student_name,
    sub.sub_name AS subject_name,
    ROUND(CAST(g.grade AS numeric)) AS rounded_score,
    sub.max_score
FROM
    student s
JOIN
    grades g ON s.stu_id = g.stu_id
JOIN
    subject sub ON g.sub_id = sub.sub_id
ORDER BY
    s.e_name,
    sub.sub_name;
 student_name | subject_name | rounded_score | max_score
--------------+--------------+---------------+-----------
(0 rows)

postgres=#
```

4-

```
postgres=# SELECT
    e_name AS student_name,
    EXTRACT(YEAR FROM birth_date) AS birth_year
FROM
    student
ORDER BY
    student_name;
 student_name | birth_year
--------------+------------
 arwa         |       1999
 aya          |       2000
 menna        |       2001
(3 rows)

postgres=#
```

5-

```
INSERT 0 1
postgres=# INSERT INTO exam (exam_id, date)
VALUES (
    (SELECT COALESCE(MAX(exam_id), 0) + 1 FROM exam),
    NOW()
)
RETURNING date;
    date
------------
 2025-04-30
(1 row)

INSERT 0 1
postgres=#
```

6-

```
postgres=# SELECT
    s.e_name AS student_name,
    ROUND(AVG(g.grade::numeric), 2) AS average_grade,
    COUNT(*) AS number_of_exams
FROM
    grades g
JOIN
    student s ON g.stu_id = s.stu_id
WHERE
    s.stu_id = 1
GROUP BY
    s.e_name;
 student_name | average_grade | number_of_exams
--------------+---------------+-----------------
(0 rows)

postgres=#
```

7-

```
postgres=# UPDATE student
SET email = REPLACE(LOWER(email), '@gmail.com', '@iti.com')
WHERE LOWER(email) LIKE '%@gmail.com';
UPDATE 3
postgres=#
```

8-

```
postgres=# SELECT
    exam_id ,
    date AS exam_date,
    CURRENT_DATE - date AS days_difference
FROM
    exam
ORDER BY
    date DESC;
 exam_id | exam_date  | days_difference
---------+------------+-----------------
       4 | 2025-04-30 |               0
       5 | 2025-04-30 |               0
       3 | 2024-03-20 |             406
       2 | 2023-12-10 |             507
       1 | 2023-09-15 |             593
(5 rows)
```

9-

```
postgres=# SELECT
    stu_id,
    e_name AS student_name,
    email
FROM
    student
WHERE
    email NOT LIKE '%.com'
    OR email IS NULL
ORDER BY
    student_name;
 stu_id | student_name | email
--------+--------------+-------
(0 rows)

postgres=#
```

10-

```
postgres=# SELECT
     exam_id,
    TO_CHAR(date, 'MM/DD/YYYY') AS formatted_date
FROM
    exam
ORDER BY
    date;
 exam_id | formatted_date
---------+----------------
       1 | 09/15/2023
       2 | 12/10/2023
       3 | 03/20/2024
       4 | 04/30/2025
       5 | 04/30/2025
(5 rows)

postgres=#
```